

ESSENTIALS OF DATA SCIENCE

Assignment No. 2: NumPy and Pandas

Name: Samrudhi Naik

PRN: 202501070179

Division: ET2

Dataset: Paper Review Dataset

The Paper Review dataset consists of structured evaluation data related to academic paper reviews. It includes reviewer identifiers, categories, knowledge levels, and numerical grades assigned to different proposals. The dataset primarily contains categorical descriptors and numerical scores rather than full textual feedback.

Using NumPy and Pandas, the dataset is analyzed to understand reviewer behavior, grading patterns, and categorical distributions.

Problem Statements

1. Load the dataset and display the first 10 rows.

```
1  ✓ import pandas as pd
2      import numpy as np
3      from collections import Counter
4
5      # Load dataset
6      df = pd.read_csv("D:\\code\\assignments\\reviews_coupled_with_feedback.csv")
7
8      # Clean column names
9      df.columns = df.columns.str.strip()
10
11     # 1. Printing first 10 rows
12     print("\n--- First 10 Rows ---")
13     print(df.head(10))
14
```

```
--- First 10 Rows ---
   Unnamed: 0  user_analysis_id  proposal_analysis_id  ...  conflicted  grade  rank_user
0           0             u770             p059  ...         0      4.0         8.0
1           1             u770             p126  ...         0      1.4         2.0
2           2             u770             p272  ...         0      3.0         5.0
3           3             u770             p205  ...         0      3.0         5.0
4           4             u770             p202  ...         0      1.5         3.0
5           5             u770             p564  ...         0      3.0         5.0
6           6             u770             p295  ...         0      1.2         1.0
7           7             u770             p159  ...         0      3.2         7.0
8          24             u814             p201  ...         0      2.5         4.0
9          25             u814             p240  ...         0      1.5         1.0

[10 rows x 12 columns]
```

2. Check for missing values in the dataset.

```
15 # 2. Missing values
16 print("\n--- Missing Values ---")
17 print(df.isnull().sum())
18
```

```
--- Missing Values ---
user_analysis_id      0
proposal_analysis_id  0
review_group          0
sci_cat               0
ref_knowledge         56
ref_knowledge_text    56
dt_knowledge          416
dt_knowledge_group    416
conflicted            0
grade                61
rank_user             61
dtype: int64
```

3. Display data types of all columns.

```
19 # 3. Data types
20 print("\n--- Data Types ---")
21 print(df.dtypes)
22
```

```
--- Data Types ---
user_analysis_id      str
proposal_analysis_id  str
review_group          str
sci_cat               str
ref_knowledge         float64
ref_knowledge_text    str
dt_knowledge          float64
dt_knowledge_group    str
conflicted            int64
grade                float64
rank_user             float64
dtype: object
```

4. Count the total number of records.

```
23 # 4. Total records
24 print("\nTotal records:", len(df))
25
```

```
Total records: 1088
```

5. Remove unnecessary columns such as indexing columns.

```
26 # 5. Drop unnecessary column
27 if 'Unnamed: 0' in df.columns:
28     df = df.drop(columns=['Unnamed: 0'])
29
```

6. Convert the grade column into numeric format.

```
30 # 6. Converting to numeric
31 df['grade'] = pd.to_numeric(df['grade'], errors='coerce')
32
```

7. Handle missing values in the grade column.

```
33 # 7. Handling missing values in grade column
34 df = df.dropna(subset=['grade'])
35
```

8. Calculate the average grade of all reviews.

```
36 # 8. Average of grade of all reviews
37 grades = df['grade'].values
38 print("\n--- Average grade ---")
39 print("Mean:", np.mean(grades))
40
```

```
--- Average grade ---
Mean: 2.397809152872444
```

9. Compute median, and standard deviation of grades using NumPy.

```
41 # 9. Median and Standard Deviation of all reviews
42 print("Median:", np.median(grades))
43 print("Std Dev:", np.std(grades))
44
```

```
--- Median and Standard Deviation ---
Median: 2.4
Std Dev: 0.7615310071001948
```

10. Identify maximum and minimum grades.

```
46 # 10. Maximum and Minimum marks
47 print("\nMax grade:", df['grade'].max())
48 print("Min grade:", df['grade'].min())
49
```

```
Max grade: 5.0
Min grade: 1.0
Variance: 0.579929474775037
```

11. Calculate the variance of grades.

```
50 # 11. Variance of all reviews
51 print("Variance:", np.var(grades))
52
```

```
--- Variance ---
Variance: 0.579929474775037
```

12. Find the average grade for each review group.

```
54 # 12. Average grade for each review group
55 print("\n--- Avg Grade per Review Group ---")
56 print(df.groupby('review_group')['grade'].mean())
57
```

```
--- Avg Grade per Review Group ---
review_group
deephought      2.371313
opc_emulate     2.437683
Name: grade, dtype: float64
```

13. Find the average grade for each scientific category.

```
58 # 13. Average grade for each scientific category
59 print("\n--- Avg Grade per Category ---")
60 print(df.groupby('sci_cat')['grade'].mean())
61
```

```
--- Avg Grade per Category ---
sci_cat
A      2.513278
B      2.398826
C      2.397034
D      2.315134
Name: grade, dtype: float64
```

14. Count the number of reviews per review group.

```
62 # 14. Number of reviews per review group
63 print("\n--- Count per Review Group ---")
64 print(df['review_group'].value_counts())
65
```

```
--- Count per Review Group ---
review_group
deephought      617
opc_emulate     410
Name: count, dtype: int64
```

15.Filter records where grade is above the average.

```
66 # 15. Filtering grade above mean grade
67 avg_grade = df['grade'].mean()
68 print("\n--- High Grade Records ---")
69 print(df[df['grade'] > avg_grade].head())
70
```

```
--- High Grade Records ---
  user_analysis_id proposal_analysis_id review_group ... conflicted grade rank_user
0             u770             p059  deeptthought ...           0     4.0         8.0
2             u770             p272  deeptthought ...           0     3.0         5.0
3             u770             p205  deeptthought ...           0     3.0         5.0
5             u770             p564  deeptthought ...           0     3.0         5.0
7             u770             p159  deeptthought ...           0     3.2         7.0
```

16.Identify records with grade below the average.

```
71 # 16. Filtering grade below mean grade
72 print("\n--- Low Grade Records ---")
73 print(df[df['grade'] < avg_grade].head())
74
```

```
--- Low Grade Records ---
  user_analysis_id proposal_analysis_id review_group ... conflicted grade rank_user
1             u770             p126  deeptthought ...           0     1.4         2.0
4             u770             p202  deeptthought ...           0     1.5         3.0
6             u770             p295  deeptthought ...           0     1.2         1.0
9             u814             p240  deeptthought ...           0     1.5         1.0
11            u814             p202  deeptthought ...           0     2.2         3.0
```

17.Count the number of conflicted and non-conflicted reviews

```
75 # 17. Counting number of conflicted and non-conflicted reviews
76 print("\n--- Conflicted Count ---")
77 print(df['conflicted'].value_counts())
78
```

```
[5 rows x 11 columns]

--- Conflicted Count ---
conflicted
0      1025
1         2
Name: count, dtype: int64
```

18.Create a new column for word count in feedback text.

```
79 # 18. Creating new column for word count
80 df['ref_knowledge_text'] = df['ref_knowledge_text'].fillna("")
81 df['word_count'] = df['ref_knowledge_text'].str.split().str.len()
82
```

19. Analyze the distribution of knowledge categories in feedback.

```
83 # 19. Analysing distribution of knowledge categories
84 print("\nAverage word count:", df['word_count'].mean())
85
```

```
Average word count: 1.0
```

20. Identify the most frequent feedback terms.

```
86 # 20. Identifying most frequent feedback terms
87 words = " ".join(df['ref_knowledge_text']).lower().split()
88 freq = Counter(words)
89
90 print("\n--- Most Common Words ---")
91 print(freq.most_common(10))
```

```
--- Most Common Words ---
[('little_knowledge', 392), ('general_knowledge', 391), ('expert', 244)]
```

Conclusion

The Paper Review dataset was successfully analyzed using NumPy and Pandas to extract insights into grading patterns and reviewer characteristics. The analysis revealed moderate grading trends, minimal conflict in reviews, and a dominance of general knowledge-level reviewers. The dataset demonstrates how structured and categorical data can be effectively analyzed using data science tools.